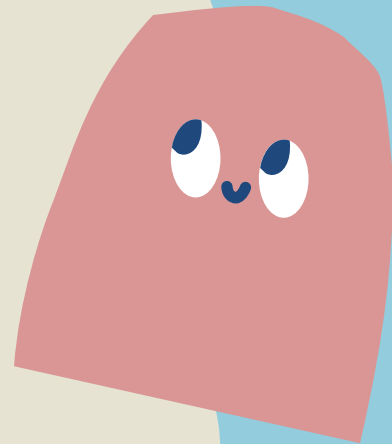
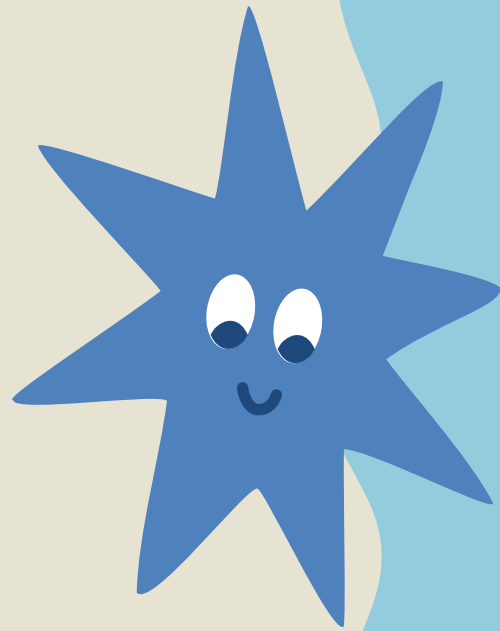


Syntax Analysis in NLP



Syntax Analysis

- Syntax Analysis is the process of analyzing the grammatical structure of a sentence.
- It determines how words are arranged and related.
- Also known as **Parsing**.
- Ensures that a sentence follows the grammar rules of a language.

Example

Sentence: *"The cat chased the mouse."*

- Subject → The cat
- Verb → chased
- Object → the mouse
- **Purpose:** Convert text into a structured representation for machines.

Types of Syntax Analysis

1. Constituency Parsing

- Breaks a sentence into phrase structures.
- Creates a **Parse Tree**.
- Groups words into noun phrases (NP), verb phrases (VP), etc.
- Example:
(S
 (NP The cat)
 (VP chased
 (NP the mouse)))

2. Dependency Parsing

- Shows grammatical relationships between words.
- Connects each word to its governing word.
- Produces a Dependency Tree.
- Example:
chased
├── cat (subject)
└── mouse (object)

Challenges & Summary

Challenges

- Ambiguous sentence structures
- Complex grammar rules
- Long and nested sentences
- Language variations and idioms

Example of Ambiguity

- "I saw the man with the telescope."
- Did **I** use the telescope?
- Or **did the man** have the telescope?

Summary

- Syntax Analysis identifies grammatical structure.
- Uses **Constituency Parsing** and **Dependency Parsing**.
- Plays a vital role in modern NLP applications such as translation, chatbots, and search engines.

THANK YOU